



The Autostar/LX200 Command Set and Differences

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Organization of Commands

The grammar for the LX200 command set is organized according to the action the telescope takes in response to the command.

There are twenty categories, or groups, of commands according to this 'action' view. The groups are shown in the table below.

Telescope Command Groupings:

Command Group	Command Designator Symbol
Alignment Query	<ACK>
Alignment	A
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Some of the commands are specific to the LX200 16-inch telescopes. These commands are marked in the above table using an asterisk following the Command Description on the left hand side of the table.

Each command grouping may have sub-groupings. The Library command group, as an example, can be further broken down into the logical categories of Library Commands affecting the Telescope and Library Commands which affect Objects.

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Alphabetical List of Commands

[<ACK> A B C D f* F G h* H L M P \\$Q* Q r*R S T U V* W* X* ?](#)

Notes:
* next to a command identifier indicates a command supported only on the LX200 16-inch telescopes. The asterisk is not a part of the command syntax.

<none> means nothing is returned from the telescope if listed in the Value(s) Returned column.
<none> means no parameter is supplied and no parameter is needed if listed in the Takes As Parameter(s) column.

Letters & digits shown are the ASCII symbols of the returned values. For example the symbol 'A' as a "Values(s) Returned" entry means that the program should expect to receive an ASCII 'A' (0x41) as the return value. Similarly a lower case 'a' would indicate that the program should expect to receive an ASCII 'a' (0x61) as the return value.

<bool> is a boolean returned value of either an ASCII '0' (0x30) or an ASCII '1' (0x31). Zero is false.

<string&62; is a sequence of characters followed by a hash-mark (ASCII '#' 0x23).

If a symbol is indicated with enclosing symbols (as in parenthesis marks around the 12 or 24 in the # :G1# command) the enclosing symbols are literal and are a part of the returned value.

A single ASCII digit as in the return value 2<string> for the command # :MS# indicates an ASCII '2' (0x32) value is prefixed to the <string>. The ASCII digit and the <string> are both part of the returned value.

ACK (Autostar does not return a 'G' ever.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
ACK<ASCII 6>				Alignment Query
		<none>	A	AltAz
			G	German Polar Mount
			L	Land
			P	Polar

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A (Autostar)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
A				Set Alignment Mode
	L	<none>	<none>	Land
	P	<none>	<none>	Polar
	A	<none>	<none>	AltAz

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B (Autostar) [pending]

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
B				Reticle Brightness Control
	+	<none>	<none>	Increase
	-	<none>	<none>	Decrease
	0	<none>	<none>	Flashing Mode 0
	1	<none>	<none>	Flashing Mode 1
	2	<none>	<none>	Flashing Mode 2
	3	<none>	<none>	Flashing Mode 3

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C (Autostar always returns the same string of:" M31 EX GAL MAG 3.5 SZ178.0 '# " (34 ASCII characters)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
C				Sync -- Match Scope Coord to Object Coords
	M	<none>	<string>	Which Object's coordinates were used.

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D (Autostar returns a single bar if continuing to slew and otherwise an empty string is returned.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
D				Get Distance Bars String
		<none>	<string>	Number of bars on the display (see exceptions)

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f (Autostar)

Command (Major	Command (Minor	Takes As Parameter(s)	Value(s) Returned	Description
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Grouping)	Grouping)			
f				Fan Control (16-inch only)
	+	<none>	<none>	On
	-	<none>	<none>	Off

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F (Autostar supports 4 speeds) [pending]

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
F				Focus Control
	+	<none>	<none>	In
	-	<none>	<none>	Out
	Q	<none>	<none>	Quit Focus Command
	F	<none>	<none>	Focus Fast
	S	<none>	<none>	Focus Slow

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G (Autostar: See [Autostar GET Command](#) section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
G				Get Information
	0	<none>	Alignment Menu 0#	Alignment Menu Entry 0
	1	<none>	Alignment Menu 1#	Alignment Menu Entry 1
	2	<none>	Alignment Menu 2#	Alignment Menu Entry 2
	A	<none>	sDD*MM#	Telescope Altitude
	a	<none>	HH:MM:SS#	Local Time in 12-hour Format
	b	<none>	sMM.M#	Brighter
	C	<none>	MM/DD/YY#	Calendar Date
	c			Clock 12 or 24-hour status
		<none>	(12)#	
		<none>	(24)#	

	D	<none>	sDD*MM#	Telescope Declination
	d	<none>	sDD*MM#	Object Declination
	F	<none>	NNN'#	Field
	f	<none>	sMM.M#	Fainter Limit
	G	<none>	sHH#	Hours From UCT (GMT)
	g	<none>	DDD*MM#	Site Longitude
	h	<none>	DD*#	Higher Limit
	L	<none>	HH:MM:SS#	Local Time in 24-hour Format
	l	<none>	NNN'#	Larger Size Limit
	M	<none>	XYZ#	Site Name 1
	N	<none>	XYZ#	Site Name 2
	o	<none>	DD*#	Lower limit
	O	<none>	XYZ#	Site Name 3
	P	<none>	XYZ#	Site Name 4
	q			Minimum Quality For Find Operation
		<none>	SU#	Super
		<none>	EX#	Excellent
		<none>	VG#	Very Good
		<none>	GD#	Good
		<none>	FR#	Fair
		<none>	PR#	Poor
		<none>	VP#	Very Poor
	R	<none>	+HH:MM.T#	Telescope RA
	r	<none>	HH:MM.T#	Object RA
	S	<none>	HH:MM:SS#	Sidereal Time (24-hour)
	s	<none>	NNN'#	Smaller Size Limit
	T	<none>	TT.T#	Current Track Frequency
	t	<none>	sDD*MM#	Site Latitude
	y	<none>	GPDCO#	Types Parameters (Capital == Selected else not)

	Z	<none>	DDD*MM#	Telescope Azimuth
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h (Autostar obeys the Park minor command and returns nothing on all other 'h' commands.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
h				Home Position Search Controls (16" only)
	S	<none>		Start Home Position Search, Save Current Position
	F	<none>		Start Home Position Search Set To Saved Values
	P	<none>		Slew to Home Position
	?			
		<none>	0	Home Search Failed
		<none>	1	Home Search Found
		<none>	2	Home Search in Progress

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H (Autostar always uses an internal 24-hour clock and ignores this command)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
H		<none>	<none>	Toggles 12 or 24-Hour Time Format

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L (Autostar: See [Autostar Library Command](#) Section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
L				Library Control

	B	<none>	<none>	Previous Object In Find
	C	NNNN	<none>	Find CNGC Type Objects
	F	<none>	<none>	Start Find Operation
	f			Field
		<none>	<string>	Number of objects in field & object in center field.
	I	<none>	<string>	Object Information
	M	NNNN	<none>	Find Messier
	N	<none>	<none>	Next Object In Find
	o			Set the NGC library type
		0	<bool>	Objects CNGC
		1	<bool>	Objects IC
		2	<bool>	Objects UGC
	s			Set the STAR object library
		0	<bool>	STAR library
		1	<bool>	SAO library
		2	<bool>	GCVS library
	S	NNNN	<none>	Find STAR Type Objects

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M (Autostar: See [Autostar Movement Command](#) Section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
M				Move
	A	<none>	0	Slew to AltAz (16-inch only) used with Sa Sz cmd
	e	<none>	<none>	East
	n	<none>	<none>	North
	s	<none>	<none>	South
	w	<none>	<none>	West

	S			Slew to Object
		<none>	0	Slew is Possible
		<none>	1<string>	Object Below Horizon w/string message
		<none>	2<string>	Object Below Higher w/string message

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P (Autostar always returns "HIGH PRECISION". The Autostar only knows one precision and does not switch.")

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
P				High Precision
	<none>	<none>	"HIGH PRECISION"	When On (14 character string)
	<none>	<none>	"LOW PRECISION"	When Off (14 character string)

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\$Q (Autostar ignores)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
\$Q				Toggles Smart Drive Status
	1	<none>	<none>	
	2	<none>	<none>	
	3	<none>	<none>	
	4	<none>	<none>	
	5	<none>	<none>	

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Q (Autostar)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description

Q				Abort Movement
	e	<none>	<none>	East
	n	<none>	<none>	North
	s	<none>	<none>	South
	w	<none>	<none>	West
	<none>	<none>	<none>	Stop the Slew

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R (Autostar ignores)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
r				Field De- Rotater (16" only)
	+	<none>	<none>	On
	-	<none>	<none>	Off

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R (Autostar: See Autostar Rate Command Section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
R				Set Movement Rate
	C	<none>	<none>	Center
	G	<none>	<none>	Guide
	M	<none>	<none>	Move
	S	<none>	<none>	Slew

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S (Autostar: See [Autostar SET Command](#) Section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
S				Set Telescope or Object Parameter

	a	sDD*MM	<bool>	Object Altitude (16-inch only) used with MA cmd
	b	sMM.M	<bool>	Brighter
	C	MM/DD/YY	<bool><string>	Calendar: If the date is valid 2 <string>s are returned, each string is 31 bytes long. The first is: "Updating planetary data#" followed by a second string of 30 spaces terminated by '#'
	d	sDD*MM	<bool>	Object DEC
	f	sMM.M	<bool>	Fainter
	F	NNN	<bool>	Field
	g	DDD*MM	<bool>	Site Longitude
	G	sHH	<bool>	Hours From UTC (GMT)
	h	DD	<bool>	Higher
	l	NNN	<bool>	Larger
	L	HH:MM:SS	<bool>	Local
	M	XYZ	<bool>	Site Name 1
	N	XYZ	<bool>	Site Name 2
	o	DD*	<bool>	Lower Limit
	O	XYZ	<bool>	Site Name 3
	P	XYZ	<bool>	Site Name 4
	q	<none>	<none>	Quality
	r	HH:MM.T	<bool>	Object RA
	s	NNN	<bool>	Smaller
	S	HH:MM:SS	<bool>	Sidereal
	t	sDD*MM	<bool>	Site Latitude
	T	TT.T	<bool>	Frequency
	w	N	<bool>	Max Slew Rate dgrees/second (2..4)

	y	GPDCO	<bool>	Find String Type: Capitals indicate active
	z	DDD*MM	<bool>	Object Azimuth (16-inch only) used with MA cmd

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T (Autostar: See [Autostar TRACKING Command](#) Section.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
T				Tracking Frequency
	M	<none>	<none>	Switch to Manual
	Q	<none>	<none>	Switch to Quartz
	+	<none>	<none>	Increment by one tenth
	-	<none>	<none>	Decrement by one tenth

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U (Autostar)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
U	<none>	<none>	<none>	Toggles Long format on and off.

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V (Autostar ignores)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
V				View PEC Scale Data (16" only)

	D	NNN	N.NNNNNN#	DEC Scale Data
	R	NNN	N.NNNNNN#	RA Scale Data

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W (Autostar will set User sites 0..3, depending on the internal variable MAXUSERSITE. If site > MAXUSERSITE, command is ignored.)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
W				Set Waypoint (Site) Number
	1	<none>	<none>	Set To Site 1
	2	<none>	<none>	Set To Site 2
	3	<none>	<none>	Set To Site 3
	4	<none>	<none>	Set To Site 4

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X (Autostar ignores)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
X				Worm Sector Position (16" only)
	D	<none>	<none>	DEC ALT Current Worm Sector Position
	R	<none>	<none>	RA AZ Current Worm Sector Position
	Y			Write Targeted Sector
		D	<none>	Write to DEC EEPROM
		R	<none>	Write to RA EEPROM
	Z	<none>	<none>	Target Sector of a worm (0..199)

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? (Autostar ignores)

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
?				Help Menu Page Strings
	?	<none>	<string>	Starts return of the page
	+	<none>	<string>	Next page
	-	<none>	<string>	Previous page

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Autostar GET Commands

Command (Major Grouping)	Command (Minor Grouping)	Autostar Differences	Value(s) Returned	Description
G				Get Information
	0	<empty-string>	Alignment Menu 0#	Alignment Menu Entry 0
	1	<empty-string>	Alignment Menu 1#	Alignment Menu Entry 1
	2	<empty-string>	Alignment Menu 2#	Alignment Menu Entry 2
	A	<none>	sDD*MM#	Telescope Altitude
	a	<none>	HH:MM:SS#	Local Time in 12-hour Format
	b	<none>	sMM.M#	Brighter
	C	<none>	MM/DD/YY#	Calendar Date
	c			Clock 12 or 24-hour status
		<never done>	(12)#	
		<none>	(24)#	
	D	<none>	sDD*MM#	Telescope Declination
	d	<none>	sDD*MM#	Object Declination
	F	<none>	NNN'#	Field
	f	<none>	sMM.M#	Fainter Limit
	G	<none>	sHH#	Hours From UCT (GMT)
	g	<none>	DDD*MM#	Site Longitude

	h	<none>	DD*#	Higher Limit
	L	<none>	HH:MM:SS#	Local Time in 24-hour Format
	l	<none>	NNN' #	Larger Size Limit
	M	<16-char-name>	XYZ#	Site Name 1
	N	<16-char-name>	XYZ#	Site Name 2
	o	<nonr>	DD*#	Lower limit
	O	<16-char-name>	XYZ#	Site Name 3
	P	<16-char-name>	XYZ#	Site Name 4
	q	Has no meaning to Autostar, but is processed by Autostar LX-200 command set.		Minimum Quality For Find Operation
		<none>	SU#	Super
		<none>	EX#	Excellent
		<none>	VG#	Very Good
		<none>	GD#	Good
		<none>	FR#	Fair
		<none>	PR#	Poor
		<none>	VP#	Very Poor
	R	<none>	HH:MM.T#	Telescope RA
	r	<none>	HH:MM.T#	Object RA
	S	<none>	HH:MM:SS#	Sidereal Time (24-hour)
	s	<none>	NNN' #	Smaller Size Limit
	T	<none>	TT.T#	Current Track Frequency
	t	<none>	sDD*MM#	Site Latitude
	V	(NOT in LX200 Command Set)		Get Version Information
		D	<string>	Version Date
		F	<string>	Full Version String
		N	<string>	Version Number (major.minor)
		P	<string>	Product Name

		T	<string>	Version Time
	y	Has no meaning, but is processed by Autostar LX-200 command set <none>	GPDCO#	Types Parameters (Capital == Selected else not)
	Z	<none>	DDD*MM#	Telescope Azimuth

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Autostar Library Commands

Library Commands are generally not processed by the Autostar, because the LX-200 Libraries and the Autostar databases are so different. Some commands (see :Lo# and Ls# are tracked, that is, from the LX-200 command set, it will appear that the command happened, but actually there is no change in the Autostar at all.

Command (Major Grouping)	Command (Minor Grouping)	Autostar Differences	Value(s) Returned	Description
L				Library Control
	B	<ignored>	<none>	Previous Object In Find
	C	<ignored>	<none>	Find CNGC Type Objects
	F	<ignored>	<none>	Start Find Operation
	f			Field
		<always>	<string>	"objects: 0 #"
	I	<always>	<string>	" M31 EX GAL MAG 3.5 SZ178.0'#"
	M	<ignored>	<none>	Find Messier
	N	<ignored>	<none>	Next Object In Find
	o	Does not mean anything to the Autostar, but is tracked.		Set the NGC library type
		<none>	<bool>	Objects CNGC
		<none>	<bool>	Objects IC
		<none>	<bool>	Objects UGC
	s	Does not mean anything to the Autostar, but is tracked.		Set the STAR object library
		<none>	<bool>	STAR library

		<none>	<bool>	SAO library
		<none>	<bool>	GCVS library
	S	<ignored>	<none>	Find STAR Type Objects

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Autostar Move Commands

Command (Major Grouping)	Command (Minor Grouping)	Takes As Parameter(s)	Value(s) Returned	Description
M				Move
	A	<none>	0	Slew to AltAz (16-inch only) used with Sa Sz cmd
	e	<none>	<none>	East
	n	<none>	<none>	North
	s	<none>	<none>	South
	w	<none>	<none>	West
	S			Slew to Object
		<none>	0	Slew is Possible
		<none>	1<string>	Object Below Horizon w/string message
		<never done>	2<string>	Object Below Higher w/string message

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Autostar Set Commands

The command most specifically requested in the LX-200 command set for the Autostar is the :Sw<N># command where <N> is any number between 2 and 4. This command is the same as pressing the 1, 5 or 9 keys on the keypad to control the maximum slew speed.

Command (Major Grouping)	Command (Minor Grouping)	Autostar Differences	Value(s) Returned	Notes
S				Set Telescope or Object Parameter
	a	<none>	<bool>	Object Altitude (16-inch only) used with MA

				cmd
	b	<none>	<bool>	Brighter
	C	The string outputs follow almost immediately from the Autostar, which does not happen on the LX-200.	<bool><string>	Calendar: If the date is valid 2 <string>s are returned, each string is 31 bytes long. The first is: "Updating planetary data#" followed by a second string of 30 spaces terminated by '#'
	d	<none>	<bool>	Object DEC
	f	<none>	<bool>	Fainter
	F	<none>	<bool>	Field
	g	<none>	<bool>	Site Longitude
	G	<none>	<bool>	Hours From UTC (GMT)
	h	<none>	<bool>	Higher
	l	<none>	<bool>	Larger
	L	<none>	<bool>	Local
	M	<16-character-string>	<bool>	Site Name 1 is set to the string
	N	<16-character-string>	<bool>	Site Name 2
	o	<none>	<bool>	Lower Limit
	O	<16-character-string>	<bool>	Site Name 3
	P	<16-character-string>	<bool>	Site Name 4
	q	<none>	<none>	Quality
	r	<none>	<bool>	Object RA
	s	<none>	<bool>	Smaller
	S	<none>	<bool>	Sidereal

	t	<none>	<bool>	Site Latitude
	T	<none>	<bool>	Frequency
	w	<none>	<bool>	Max Slew Rate dgrees/second (2..4). The Autostar has no equivalent to this command physically. So the code simply maps the input as follows: 2 == slowest (key 0) 3 == medium (key 5) 4 == fastest (key 9)
	y	<none>	<bool>	Find String Type: Capitals indicate active
	z	<none>	<bool>	Object Azimuth (16-inch only) used with MA cmd

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Autostar Track Commands

None of the Tracking commands have equivalent meaning in the Autostar. All tracking commands are ignored.

Command (Major Grouping)	Command (Minor Grouping)	Autostar Differences	Value(s) Returned	Description
T				Tracking Frequency
	M	<ignored>	<none>	Switch to Manual
	Q	<ignored>	<none>	Switch to Quartz
	+	<ignored>	<none>	Increment by one tenth
	-	<ignored>	<none>	Decrement by one tenth

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Grammar

Special Non-terminal Characters

There are four special non-terminal characters in the grammar. The **<command-prefix>** non-terminal (see the BNF or RR Syntax pages) is a hash-mark (or number sign) character (ASCII 0x23 '#') followed by a colon character (ASCII 0x3a ':'). The **<command-suffix>** non-terminal is a hash-mark (or number sign) character (ASCII 0x23 '#'). An ACK non-terminal (ASCII 0x06) is also used as a Command in the grammar (returning the current Alignment mode). And finally the degree symbol (ASCII 223 '°' [0xdf]) non-terminal is used as the separator character in degrees and minutes. This character is shown as an asterisk (ASCII 0x2a '*') in the manual for both the 16-inch and smaller telescopes and is followed by a note in the manual describing the ASCII value of the character. On the hand controller, however, the value appears as a degree symbol.

The grammar's syntax is simple consisting of the command prefix of a hash-mark and colon pair (ASCII 0x23 "#", ASCII 0x3A ":"), followed by a **<statement>**, followed by an end-character of a hash (or number) mark (ASCII 0x23 "#"). Making the **<command>** production rule:

```
<lx200-command> <=> <command-prefix> <statement> <command-suffix>
<command-prefix> <=> '#' ':'
<command-suffix> <=> '#'
```

<statement> is defined more formally in the BNF formalism detailed in the next section. Commands are case-sensitive across the grammar. For instance **:H#** is quite different than **:h#**. The first command toggles the 12/24 hour flag in the firmware and the second should return a value indicating whether the search for home position succeeded, failed or is in progress.

Command processing begins upon receipt of the **<command-suffix>**.

The LX200 16-inch telescope has more statements than the smaller telescopes. The statements that are unique to the 16-inch telescope are indicated in the tabular listing of the commands contained in this HTML document.

NOTES:

There is some confusion as to the **<command-prefix>** of "#:". Many programmers simply use the ":" character alone as the **<command-prefix>**. This is perfectly valid. However, since there is no return value from many of the LX200 commands, there is no way to tell if the command was actually accomplished or if there was an error. Worse yet, there is no specific way to tell whether the command was even received. If the programmer begins each of the commands with a **<command-prefix>** of "#:" then the hash-mark will act as a terminator to any awaiting input. This may trigger a pending error, but it is possible to resync to the command processing state machine at this point and continue. Another method is to send a hash-mark by itself if no response has been received and listen for return characters. Either of these methods should work fine. For an example of this input methodology, look at the BASIC LX200 demonstration code provided in the back of the LX200 Instruction Manual.

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Return Values From Commands

Return values from the issued commands fall into 4 categories:

Returned Command Value Types	Notes	Terminated with a '#'	Example Command	Example Returned Value
numeric	NNNN	Sometimes		

booleans	An ASCII '0' (0x30) or an ASCII '1' (0x31). The '0' is equivalent to false. The '1' is equivalent to true.	Never		
strings	A sequence of characters. Usually less than 33 characters in length.	Always		
none	Nothing is returned at all. Don't expect or look for a value.	Never		

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Exceptions

The **ACK** command has no command prefix or suffix (no preceding hash-mark, colon and no trailing hash-mark). It is simply an ASCII 0x06. It returns a single character which is not terminated by a hash-mark.

The **"# :D#"** command returns a string of ASCII **** (0x7f) characters, terminated by a hash-mark. The count of the **** characters is the count of the bars indicated on the display of the hand controller.

The **"# :Gy#"** command always returns the characters "GPDCO" in either upper or lower case combinations indicating which is active. Upper case characters indicate active.

The **"# :GF#"**, **"# :G1#"** and **"# :Gs#"** commands return "numeric" as their value types, followed by a quote (single quotation) character (ASCII 0x27).

The **"# :MS#"** command returns one of three values, two of which may be viewed as mixed types. A single ASCII '0' (0x30) character if returned indicates that a slew is possible. If either an ASCII '1' (0x31) or an ASCII '2' (0x32) is returned, the ASCII value will be followed by a string (which is always terminated by an ASCII '#' [0x23]). The latter two examples could be seen as strictly string returns, however the return of just an ASCII '0' is not terminated by a hash-mark! This simply is an example of a mixed type return.

The **"# :P#"** command returns 14 characters. Either the string **"HIGH PRECISION"** or the string **"LOW PRECISION"**. There are two spaces between the **"w"** in **LOW** and the **"P"** in **PRECISION**. Neither string is terminated by a hash-mark.

The **"# :SC#"** command returns a **<bool>** followed by a **<string>** followed by a **<string>**. Each string is hash-mark terminated. The second **<string>** will be all ASCII blanks (0x20) followed by the single hash-mark.

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